PUSHFORWARD ALONG A FINITE MORPHISM ON CYCLES, ON THE TRUNCATED K-GROUP AND THE COMMUTING DIAGRAM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Exactness and coherence of pushforward along a finite morphism, and the cycle class of a pushed forward sheaf. Cohomology of Schemes, labels coherent-lemma-relative-affine-vanishing (proof line 3922) and coherent-proposition-proper-pushforward-coherent (proof line 3925); Same chapter chow.tex, label lemma-cycle-push-sheaf (proof line 3928, claimed at a withdrawn revision and re-established). The first shows pushforward along the affine finite morphism is exact, the second keeps coherence, and the third gives commutativity of the right hand square by identifying the cycle class of the pushed forward sheaf.
- The mutually inverse isomorphisms between cycles and the truncated K-group. Same chapter chow.tex, label lemma-cycles-k-group (proof line 3904, claimed as S14-algebra-cycles-k-group at a withdrawn revision). Provides the horizontal bijections in the diagram, which turn commutativity on the right into commutativity on the left.

Source reference. Original lemma 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Navigation pointer for \ref{coherent-lemma-relative-affine-vanishing} used inside the counted span (line 3922); the argument that consumes it is retained in the span.

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Source reference. Original proposition 02O5: sources/stacks-chow/tags.txt, source lines 3504–3504. Navigation pointer for \ref{coherent-proposition-proper-pushforward-coherent} used inside the counted span (line 3925); the argument that consumes it is retained in the span.

Source reference. Original lemma 02R6: sources/stacks-chow/pinned-original-source.tex, source lines 1992–1992. Navigation pointer for \ref{lemma-cycle-push-sheaf} used inside the counted span (line 3928); the argument that consumes it is retained in the span.

Source reference. Original lemma 02S9: sources/stacks-chow/pinned-original-source.tex, source lines 3757–3757. Navigation pointer for \ref{lemma-cycles-k-group} used inside the counted span (line 3904); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3900); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite morphism of schemes locally of finite type over (S, δ) as in Situation 02QL. Then $\pi_* : Coh(X) \to Coh(Y)$ is an exact functor which sends $Coh_{\leq k}(X)$ into $Coh_{\leq k}(Y)$ and induces homomorphisms on K_0 of these categories and their quotients. The maps of Lemma 02S9 fit into a commutative diagram*

$$\begin{array}{ccccc} Z_k(X) & \longrightarrow & K_0(Coh_{\leq k}(X)/Coh_{\leq k-1}(X)) & \longrightarrow & Z_k(X) \\ \downarrow{\scriptstyle \pi_*} & & \downarrow{\scriptstyle \pi_*} & & \downarrow{\scriptstyle \pi_*} \\ Z_k(Y) & \longrightarrow & K_0(Coh_{\leq k}(Y)/Coh_{\leq k-1}(Y)) & \longrightarrow & Z_k(Y) \end{array}$$

Proof. A finite morphism is affine, hence pushforward of quasi-coherent modules along π is an exact functor by Cohomology of Schemes, Lemma 01XC. A finite morphism is proper, hence π_* sends coherent sheaves to coherent sheaves, see Cohomology of Schemes, Proposition 02O5. The statement on dimensions of supports is clear. Commutativity on the right follows immediately from Lemma 02R6. Since the horizontal arrows are bijections, we find that we have commutativity on the left as well. □